b) $\neg \forall x \exists y\, P(x,y) \equiv \exists x \neg \exists y\, P(x,y)$ (By De Morgan's law of quantifiers)

$\equiv \exists x \forall y \neg P(x,y)$ (By De Morgan's law of quantifiers)

c) $\neg \exists y (Q(y) \wedge \forall x \neg R(x,y)) \equiv \forall y \neg (Q(y) \wedge \forall x \neg R(x,y))$ (By De Morgan's law of quantifiers) $\equiv \forall y (\neg Q(y) \vee \neg \forall x \neg R(x,y))$ (By 1st De Morgan's law)

$\equiv \forall y (\neg Q(y) \vee \exists x \neg(\neg R(x,y)))$ (By De Morgan's law of quantifiers)

$\equiv \forall y (\neg Q(y) \vee \exists x\, R(x,y))$ (By Double-negation Law)

d) $\neg \exists y (\exists x\, R(x,y) \vee \forall x\, S(x,y)) \equiv \forall y \neg (\exists x\, R(x,y) \vee \forall x\, S(x,y))$
(By De Morgan's law of quantifiers) $\equiv \forall y (\neg \exists x\, R(x,y) \wedge \neg \forall x\, S(x,y))$
(By 2nd De Morgan's law) $\equiv \forall y (\forall x \neg R(x,y) \wedge \exists x \neg S(x,y))$
(By 2 applications of De Morgan's law of quantifiers)

e) $\neg \exists y (\forall x \exists z\, T(x,y,z) \vee \exists x \forall z\, U(x,y,z)) \equiv \forall y \neg (\forall x \exists z\, T(x,y,z) \vee \exists x \forall z\, U(x,y,z))$ (De Morgan's law of quantifiers) $\equiv \forall y (\neg \forall x \exists z\, T(x,y,z) \wedge \neg \exists x \forall z\, U(x,y,z))$ (2nd De Morgan's law) $\equiv \forall y (\exists x \neg \exists z\, T(x,y,z) \wedge \forall x \neg \forall z\, U(x,y,z))$ (2 applications of De Morgan's law of quantifiers)

$\equiv \forall y (\exists x \forall z \neg T(x,y,z) \wedge \forall x \exists z \neg U(x,y,z))$ (2 applications of De Morgan's law of quantifiers)

3) Express the negations of each of these statements so that all negation symbols immediately precede predicates.

a) $\forall x \exists y \forall z\, T(x,y,z)$

$\neg \forall x \exists y \forall z\, T(x,y,z) \equiv \exists x \neg \exists y \forall z\, T(x,y,z)$ (De Morgan's law of quantifiers)

$\equiv \exists x \forall y \neg \forall z\, T(x,y,z)$ (De Morgan's law of quantifiers)

$\equiv \exists x \forall y \exists z \neg T(x,y,z)$ (De Morgan's law of quantifiers)

b) $\forall x \exists y\, P(x,y) \vee \forall x \exists y\, Q(x,y)$

$\neg(\forall x \exists y\, P(x,y) \vee \forall x \exists y\, Q(x,y)) \equiv \neg \forall x \exists y\, P(x,y) \wedge \neg \forall x \exists y\, Q(x,y)$ (2nd De Morgan's law) ~~De Morgan's law of quantifiers~~ $\equiv \exists x \neg \exists y\, P(x,y) \wedge \exists x \neg \exists y\, Q(x,y)$ (De Morgan's law of quantifiers)

$\equiv \exists x \forall y \neg P(x,y) \wedge \exists x \forall y \neg Q(x,y)$ (De Morgan's law of quantifiers)